

Bootstrap & Resampling Methods

INS-605: Data Analysis II



Content

1 Introduction to Bootstrapping

2 Bootstrapping confidence intervals

3 Bagging Method

Helpful Resources

 Confidence Intervals by Bootstrapping Approach: A Significance Review by Mokhtar et al. (2023).

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Sciences

Mokhtar *et al.* | Malaysian Journal of Fundamental and Applied Sciences, Vol. 19 (2023) 30-42

REVIEW ARTICLE

Confidence Intervals by Bootstrapping Approach: A Significance Review

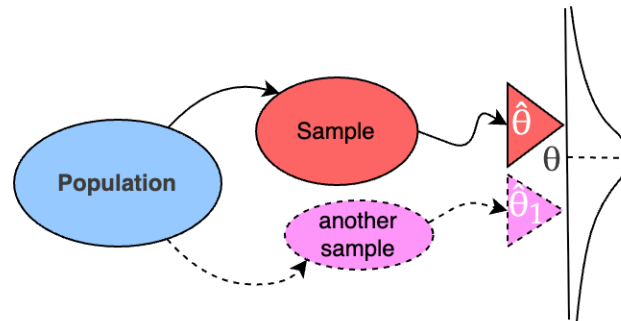
Siti Fairus Mokhtar^{a,*}, Zahayu Md Yusof^{a,c}, Hasimah Sapiri^a

^aSchool of Quantitative Sciences, Universiti Utara Malaysia 06010 Sintok, Kedah, Malaysia; ^bMathematical Sciences Studies, College of Computing, Informatics and Media, Universiti Teknologi MARA (UiTM) Kedah Branch, Sungai Petani Campus, Malaysia; ^cInstitute of Strategic Industrial Decision Modelling (ISIDM), Universiti Utara Malaysia, Kedah, Malaysia

Learning Outcomes

By the end of this lecture, you should be able to:

- Explain the process of **Bootstrapping**.
- What can we use it for in **Data Analysis & Data Science**?
- Apply **Bootstrap** to estimate **confidence intervals**.
- Apply **Bootstrap** to **Bagging**.
- Understand how correlated samples limit the power of **Bootstrap**.



1 Introduction To Bootstrapping

1.1 What Is Bootstrapping?

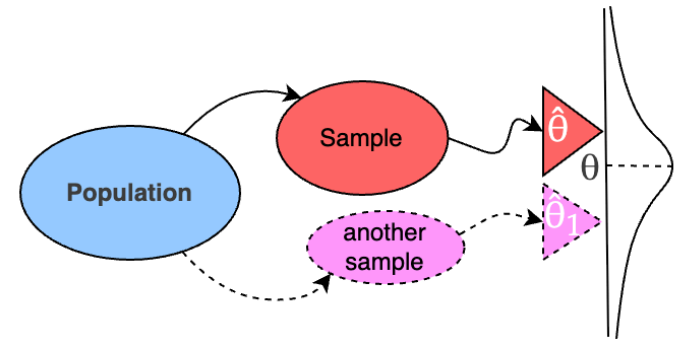
- We have only **one dataset**:

$$D = \{x_1, x_2, \dots, x_n\}$$

- But we want to know:

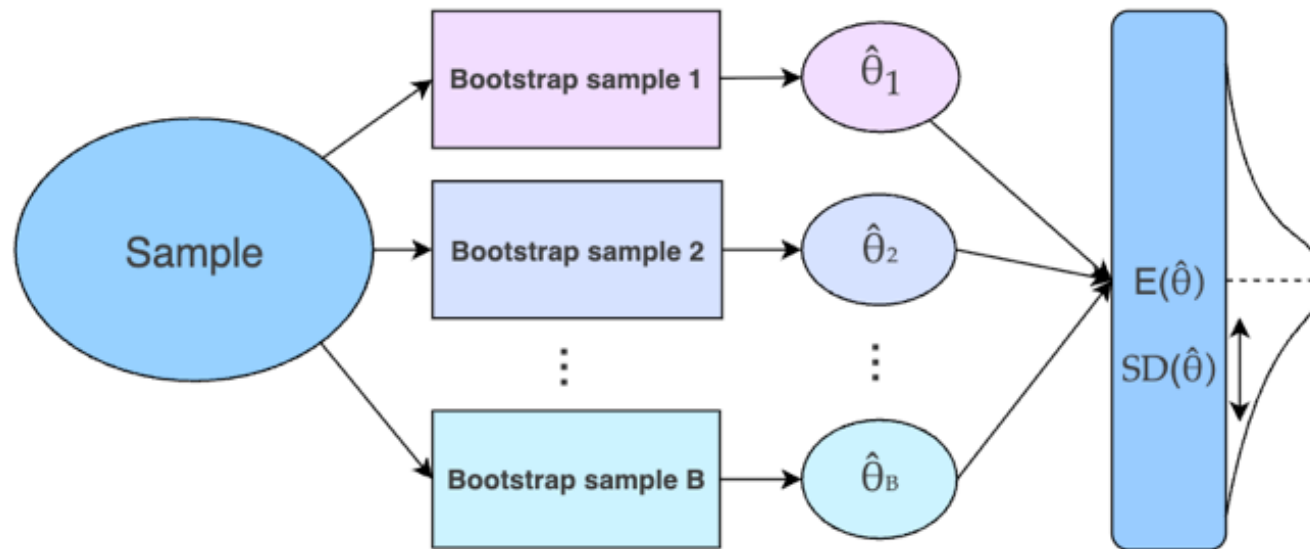
- How *variable* is our estimator?
- What would happen with *another sample*?
- How *confident* are we in our *estimate*?

- **Bootstrapping** allows us to **estimate the sampling distribution** of an estimator.
- It is a key tool for many **ML** models: **Bagging**, **RF**, etc.



1.2 Bootstrap Procedure

- Treat the observed data as an approximation of the population and repeatedly **resample with replacement**.



- Each bootstrap sample $D_j^* \rightarrow$ an estimator $\hat{\theta}_j$.
- The uncertainty of $\hat{\theta}$ is quantified by $SE(\hat{\theta})$.

2 Bootstrapping Confidence Intervals

2.1 Basic Method



Bootstrap Method

- Initialization: B (number of bootstrap samples) and the data D
 - for $b = 1, \dots, B$:
 - Draw a bootstrap sample D_b^* from the data D then compute $\hat{\theta}_b$ from D_b^* .
 - Store $\hat{\theta}_b$ to the list $\hat{\Theta}$.
 - Return $\hat{\Theta} = [\hat{\theta}_1, \dots, \hat{\theta}_B]$ and the average $\hat{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b$.
-
- Let bias $e_b = \hat{\theta}_b - \hat{\theta}^*$: the gap between estimator and the average.
 - Sort e_b 's values, then let e_q be its quantile of q -percentile.
 - Confidence interval:
$$\text{CI}_{1-\alpha}(\theta) = \left[\hat{\theta}^* - e_{\alpha/2}, \hat{\theta}^* + e_{1-\alpha/2} \right].$$

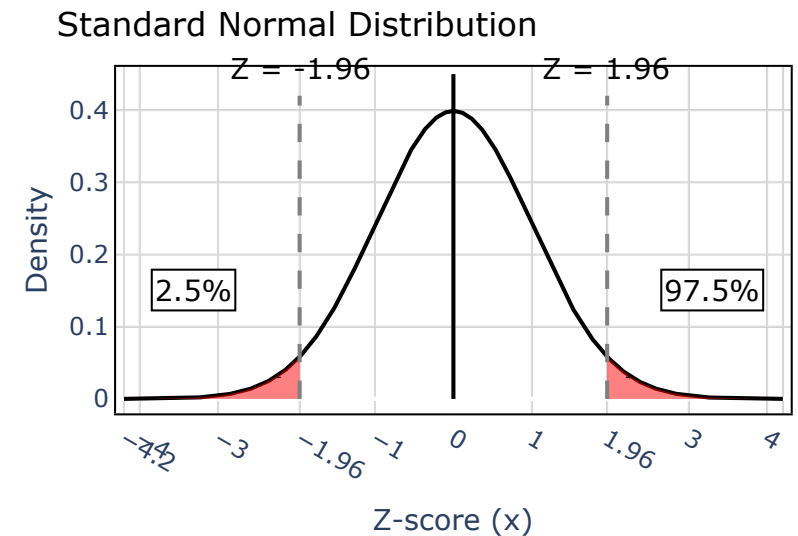
2.2 Percentile Method

💡 Bootstrap Method

- Initialization: B (number of bootstrap samples, shall be large ≈ 1000) and the data D
- for $b = 1, \dots, B$:
 - Draw a bootstrap sample D_b^* from the data D then compute $\hat{\theta}_b$ from D_b^* .
 - Store $\hat{\theta}_b$ to the list $\hat{\Theta}$.
- Return $\hat{\Theta} = [\hat{\theta}_1, \dots, \hat{\theta}_B]$ and the average $\hat{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b$.

- Sort the estimators: $\hat{\Theta} = [\hat{\theta}_{(1)}, \dots, \hat{\theta}_{(B)}]$.
- Confidence interval:

$$\text{CI}_{1-\alpha}(\theta) = [\hat{\theta}_{(\alpha/2)}, \hat{\theta}_{(1-\alpha/2)}].$$



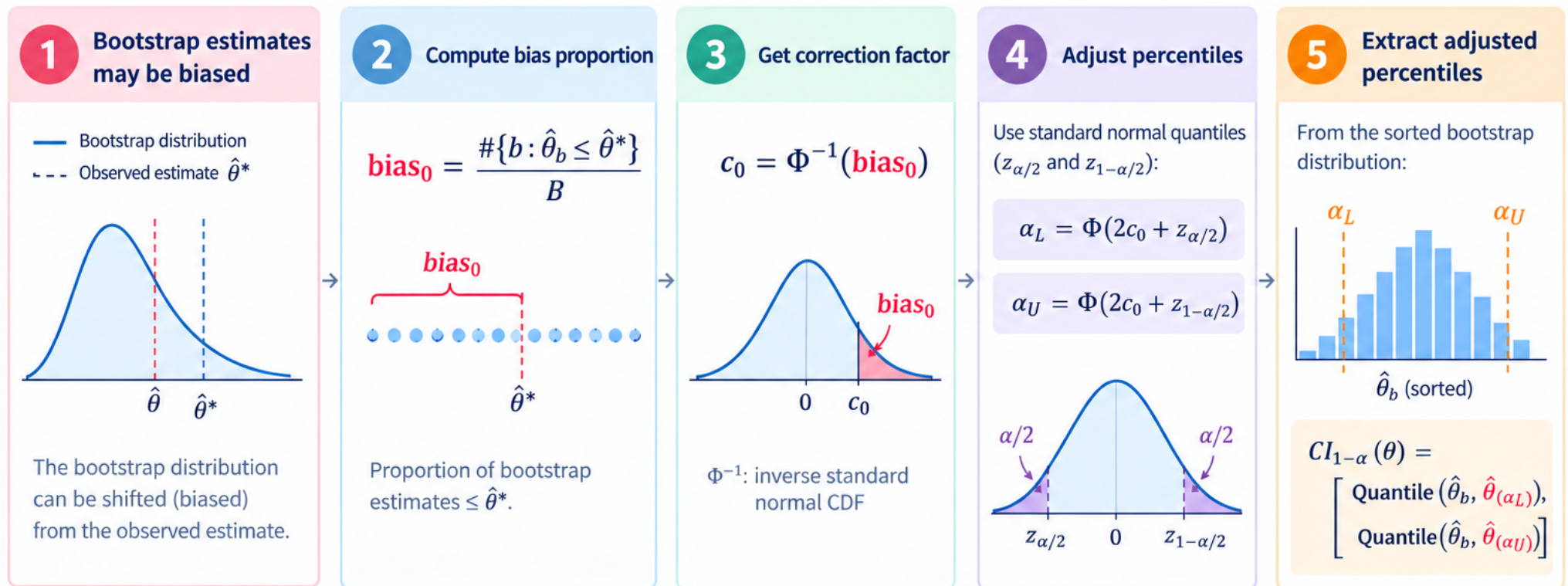
2.3 Normal Approximation Method

Bootstrap Method

- Initialization: B (number of bootstrap samples) and the data D
- for $b = 1, \dots, B$:
 - Draw a bootstrap sample D_b^* from the data D then compute $\hat{\theta}_b$ from D_b^* .
 - Store $\hat{\theta}_b$ to the list $\hat{\Theta}$.
- Return $\hat{\Theta} = [\hat{\theta}_1, \dots, \hat{\theta}_B]$ and the average & std:
$$\begin{cases} \hat{\theta}^* &= \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b \\ \text{SD} &= \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_b - \hat{\theta}^*)^2} \end{cases}$$
- Let $q_\alpha = \phi^{-1}(\alpha)$ for $\alpha \in (0, 1)$ and ϕ^{-1} is the inverse CDF of the standard normal distribution.
- Confidence interval:
$$\text{CI}_{1-\alpha}(\theta) = \left[\hat{\theta}^* - \text{SD} q_{\alpha/2}, \hat{\theta}^* + \text{SD} q_{1-\alpha/2} \right].$$

2.4 Bias-Corrected Bootstrap

- In practice, bootstrap estimates can be biased.
- **Bias-corrected bootstrap confidence interval** aims to correct this.



3 Bagging Method

3.1 From Bootstrap To Bagging

- Bootstrap is also used as the basis in **Machine Learning**.
- Train ML model f_b on a bootstrap dataset B_b :

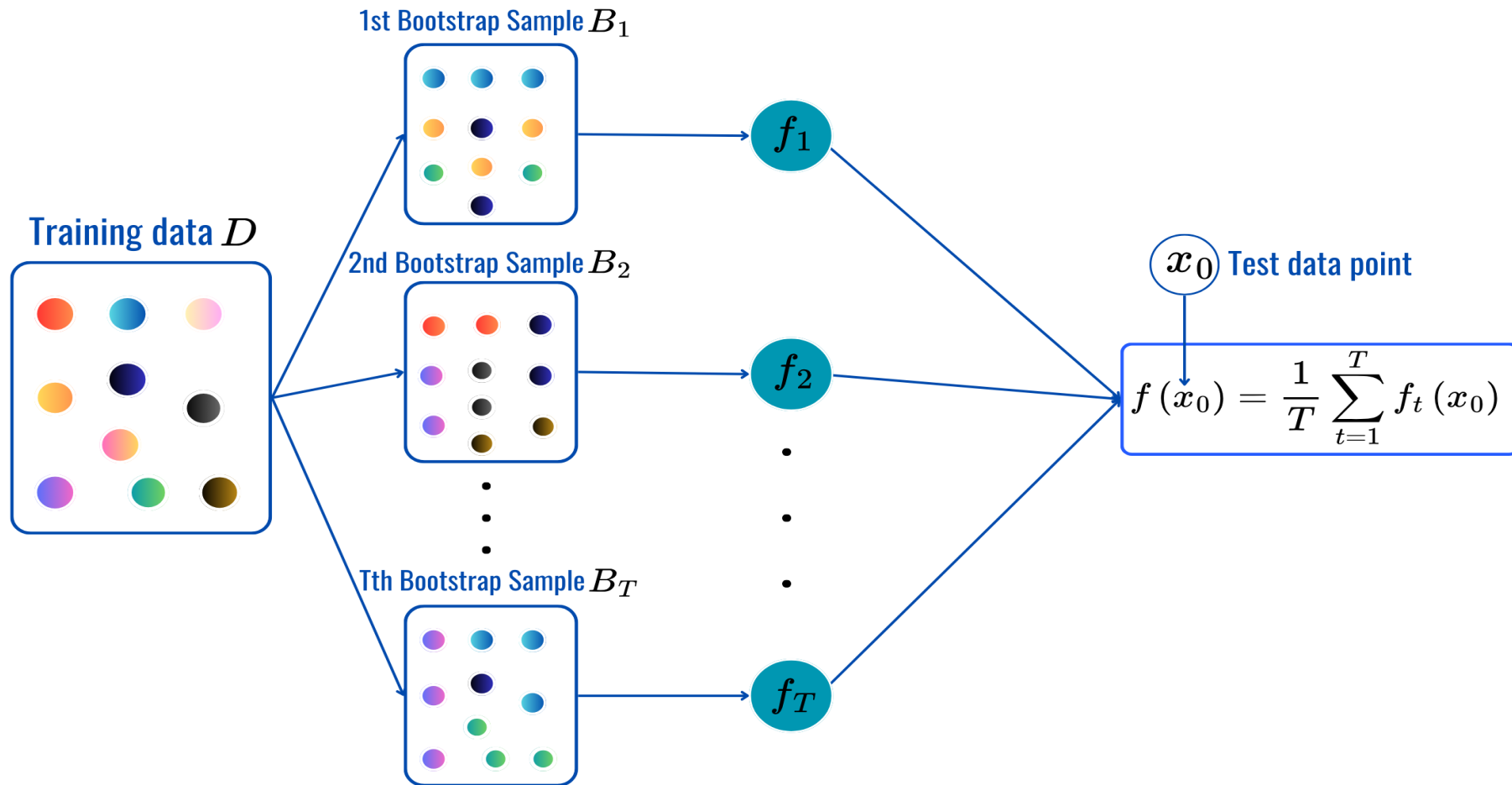
$$f_1(x), f_2(x), \dots, f_B(x)$$

- Then average their predictions:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B f_b(x)$$

- This is called: **Bootstrap Aggregating — Bagging**.

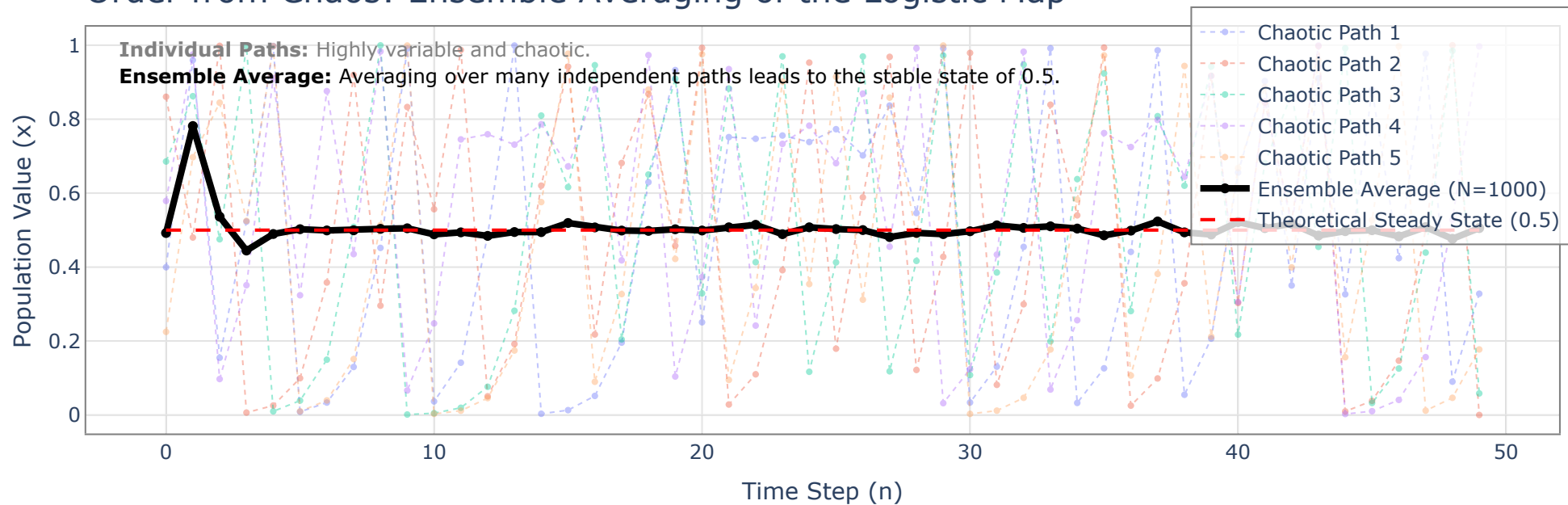
Bagging In Practice



What Does It Aim To Do?

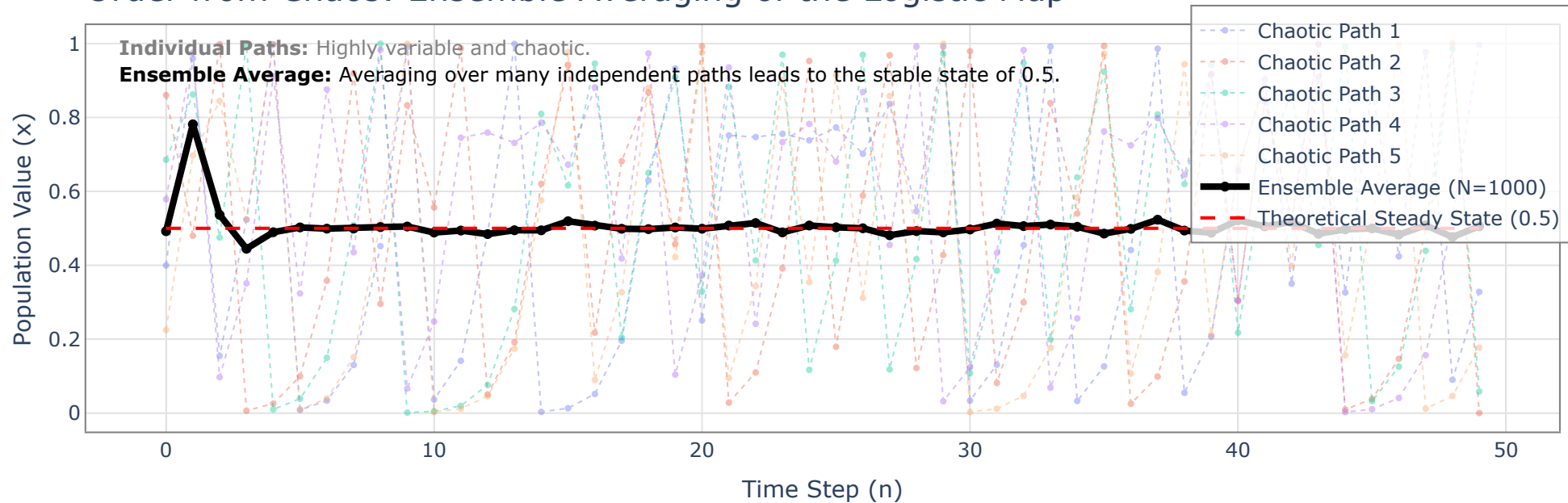
- Individual models trained on bootstrap samples can have high variance and very biased.
- Averaging several models can make the final prediction **more stable** (reducing variance).

Order from Chaos: Ensemble Averaging of the Logistic Map



4. Why Does Averaging Reduce Variance?

Order from Chaos: Ensemble Averaging of the Logistic Map



- Suppose every model has prediction variance: $\mathbb{V}(f_b) = \sigma^2$
- If models were independent: $\mathbb{V}\left(\frac{1}{B} \sum_{b=1}^B f_b\right) = \frac{\sigma^2}{B}$.
- Therefore: $B \uparrow \Rightarrow \text{Variance} \downarrow$

But Bootstrap Models Are Correlated!

- In **Bagging**, models are not actually independent!

- If pairwise correlation is ρ :
$$\mathbb{V}(\bar{f}) = \sigma^2 \left[\rho + \frac{1 - \rho}{B} \right].$$

- As B gets large: $\mathbb{V}(\bar{f}) \rightarrow \rho\sigma^2$.

- **Important ML insight:**

More models reduce variance, but **correlation creates a variance floor** (related bootstrap samples $B_b \rightarrow$ related models f_b).

Summary

- Bootstrap helps estimating uncertainty of estimators computed from different samples:
 - Standard errors
 - Several types of confidence intervals
 - Sampling distributions
 - Uncertainty of statistics
 - It's also applied to variance reduction in ML models...

🥳 Yeahhhh 🍷 !!!



Let's take a break!